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Topic : Emergency Resource Sharing

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Introduction

Emergencies can create an urgent need for essential supplies and services. People may need food, drinking water, medicines, clothes, shelter, or transportation.

At the same time, individuals, community groups, and organizations may have resources available to share. However, people in need may not know where those resources are, how much is available, or whom to contact.

The Emergency Resource Sharing System provides a common platform to publish, find, and manage available resources.

Problem Statement

During an emergency, information about available supplies is often scattered across phone calls, messages, social media posts, and local announcements. This can make it difficult to:

- Find nearby resources quickly
- Check whether a resource is still available
- Know the available quantity
- Identify the person or organization providing it
- Coordinate between people requesting help and people offering it

A centralized system can make this information easier to access and update.

Project Objectives

The main objective is to develop a web application for sharing emergency resources.

The system aims to:

- Allow providers to add resources they can share
- Display resource names, quantities, locations, and availability
- Help users browse and search for needed resources
- Let providers update or remove their listings
- Provide contact details so users can coordinate directly
- Keep resource information organized in a central database

Proposed System and Main Features

The system has two main user roles: **resource providers** and **people looking for resources**.

Providers can:

- Add a resource with its name, category, quantity, and location
- Add their contact information
- Update a listing when the quantity or availability changes
- Delete a listing when the resource is no longer available

People looking for resources can:

- View all available resources
- Search or filter listings by resource type or location
- Check the quantity and availability status
- Contact the provider using the information listed

Technology Stack and Architecture

The application uses the **MERN stack**:

- **React:** Builds the pages users interact with, including resource forms and listings
- **Node.js:** Runs the server-side application
- **Express.js:** Provides API endpoints for creating, viewing, updating, and deleting records
- **MongoDB:** Stores resource details and related records

Basic data flow:

User → React interface → Express and Node.js API → MongoDB database

When a user adds or updates a listing, the request is sent to the backend API. The backend processes it and saves the changes in MongoDB. The updated information is then shown in the React interface.

Resource Management and Workflow

Each resource record can contain:

- Resource name and category, such as food, water, medicine, or shelter
- Quantity available
- Location where the resource is available
- Provider’s name and contact details
- Availability status, such as available, limited, or unavailable

Example workflow:

1. A provider submits details about available drinking water.
2. The application sends the information to the backend.
3. The backend saves the resource record in MongoDB.
4. A person searches for water in their area and views the listing.
5. They contact the provider to coordinate collection or delivery.
6. The provider updates the quantity or availability after sharing.

Benefits, Future Scope, and Conclusion

Benefits

- Makes resource information easier to find in one place
- Helps connect providers with people who need assistance
- Makes quantities and availability visible
- Supports better coordination within a community
- Helps reduce confusion and unnecessary resource waste

Future enhancements

- Map-based resource search
- User registration and login
- Notifications when resources are added nearby
- Request and fulfilment tracking
- Admin dashboard to review listings
- Support for multiple languages

Conclusion

The **Emergency Resource Sharing System** provides a centralized platform for people to share and find essential resources during emergencies. Providers can list resources such as food, water, medicines, clothing, and shelter, along with their quantity, location, contact details, and current availability. People who need assistance can view the listings and contact providers to coordinate support.

By bringing resource information together in one place, the system can help communities respond in a more organized and timely way. Its MERN stack design supports adding, viewing, updating, and managing resource records through a user-friendly interface and a central database.

In the future, features such as map-based search, notifications, request tracking, multiple-language support, and an admin dashboard could make the platform more useful. Overall, the project offers a practical starting point for improving communication and resource coordination during emergencies.


THANK
YOU
TO ALL OF THE HEALTHCARE
& FRONTLINE WORKERS